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ANKLE

Individual fascicles of the ankle lateral ligaments and the lateral fibulotalocalcaneal ligament complex can be identified on 3D volumetric MRI

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Abstract

Purpose Lateral ligament ankle sprains are common and the anatomy on imaging studies is vital for accurate diagnosis. The lateral fibulotalocalcaneal ligament (LFTCL) complex consists of the inferior fascicle of the anterior talofibular ligament (ATFL) which is connected by arciform fibres with the calcaneofibular ligament (CFL). The superior fascicle of ATFL is an independent structure that should be assessed individually. MRI evaluation of these distinct fascicles and the arciform fibres has not been described. The aim of this study is to identify the anatomical relationship of these components of the LFTCL complex in healthy individuals on MRI.

Methods Thirty ankles from healthy volunteers were imaged using 3D volumetric MRI. The ATFL fascicles and size were evaluated. Presence of arciform fibres connecting the inferior ATFL fascicle and CFL to form the LFTCL complex and anatomical relationship around the lateral ligament complex were assessed.

Results Both the superior and inferior ATFL fascicles were observed in 26 (86.7%) ankles. The superior ATFL fascicle was significantly larger in all specimens (39% longer and 80.7% wider). For the specimens with a single fascicle, this was similar in size to the superior fascicle observed in the other 26 specimens. These measurements were not affected by age or gender. Arciform fibres of the LFTCL complex were identified in 22 (84.6%) specimens with two ATFL fascicles and three (75%) ankles with a single ATFL fascicle. Connecting fibres from the ATFL to PTFL were observed in 19 (63.3%) ankles while connections between the CFL and PTFL were identified in 21 (70%) ankles. Five ankles had a perforating artery visualized in the intervening space between the superior and inferior ATFL fascicles (a branch of the lateral tarsal artery of the dorsalis pedis artery).

Conclusion Two distinct ATFL fascicles may be identified in the majority of ankles on MRI. Isolated injury to the superior fascicle identified on MRI may be useful when diagnosing patients presenting with symptoms of subtle instability without overt ankle laxity on clinical examination. The current study is the first to identify the arciform fibres of the LFTCL complex supporting isolated ATFL repair in the presence of intact LFTCL complex.

Level of evidence Level III.

Keywords Ankle lateral ligament · MRI images · Anatomy · ATFL fascicles · Lateral fibulotalocalcaneal ligament complex · Microinstability

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Introduction

Ankle sprains are one of the most frequent injuries in sporting activities and represent 31% of all injuries in soccer and 45% of all injuries in basketball [6]. The majority are supination inversion ankle sprains affecting the lateral ligamentous complex [6, 14, 20]. This complex is formed by the anterior talofibular ligament (ATFL), the calcaneofibular ligament (CFL) and the posterior talofibular ligament (PTFL) with the ATFL most commonly injured [1, 21]. Recent anatomical studies have demonstrated that the ATFL consist of two distinct fascicles with an intra-articular superior fascicle and an extra-articular inferior fascicle [3, 4, 7, 17, 18] which is connected to the CFL via arciform fibres to form the lateral fibulotalocalcaneal ligament (LFTCL) complex functioning as an isometric lateral ankle stabilizer (Fig. 1) [18]. In addition to this, intra-articular connections between the three lateral ligaments have been recently described [4]: ATFL's inferior fascicle and CFL were found to be connected in all cases, as were CFL and PTFL; ATFL's inferior fascicle and PTFL were connected also in all cases, while ATFL's superior fascicle and PTFL were connected in 42.5% of cases.

It has been hypothesized that isolated tears of the superior ATFL fascicle, being intra-articular and bathed in synovial fluid, are unable to heal which may then lead to subtle lateral ankle instability that is difficult to recognize on clinical examination [3, 16–19]. Several studies have suggested that this isolated injury to the superior fascicle of ATFL can lead to chronic anterolateral ankle pain known as "microinstability" [3, 16–19]. Vega and colleagues described four injury patterns to the superior fascicle identified during arthroscopy in 232 patients diagnosed with ankle microinstability: loss of tension (type I); partial detachment (type II); total detachment (type III); total or partial resorption of the superior fascicle (type IV) [17].

Understanding the distinct anatomical characteristics of lateral ligamentous complex is crucial when diagnosing patients with ankle pain and ankle instability. Identifying specific injury patterns to individual fascicles of the ATFL on magnetic resonance imaging (MRI) may aid in the diagnosis of those with symptoms from subtle instability. The arciform fibre connection between the inferior fascicle of the ATFL to the CFL is vital for the success of isolated repair of the ATFL during surgery for lateral ligament instability [3, 9, 10, 18]. There are few imaging studies describing the distinct fascicles of the ATFL and none describing the characterization of the LFTCL complex on magnetic resonance imaging (MRI).

The aim of this pilot study is to identify and describe the distinct fascicles of the ATFL and the LFTCL complex (inferior fascicle of ATFL, arciform fibres and CFL) on MRI in healthy individuals. These distinct structures were hypothesized to be identifiable on MRI scan of the ankle.

Materials and methods

Approval to undertake the study was provided by the West Midlands—Coventry & Warwickshire Research Ethics Committee (Reference: 19/WM/0383). This prospective pilot study recruited 30 healthy volunteers without a history of ankle injury to undergo MRI scan of one of their ankles in

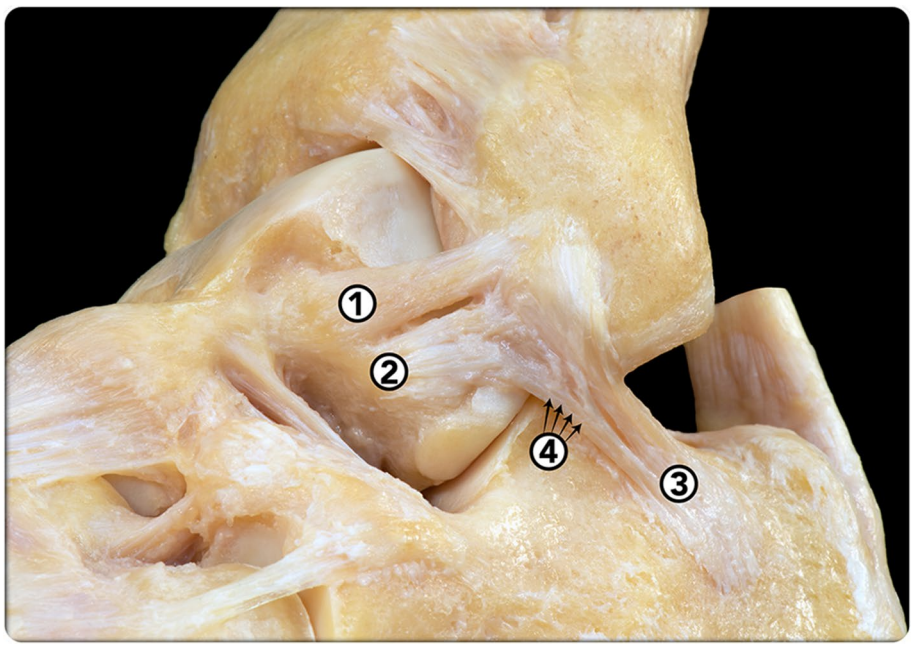


Fig. 1 Anatomical dissection showing the lateral fibulotalocalcaneal complex: 1. ATFL's superior fascicle. 2. ATFL's inferior fascicle. 3. CFL. 4. Arciform connecting fibres between ATFL's inferior fascicle and CFL

2021. Anyone with foot or ankle pain requiring modification of activities which lasted for more than one week or had history of surgery, fracture or bone stress reaction to the foot or ankle, ankle instability symptoms in the last six months were excluded. Those with previous surgery to the foot and ankle or had contraindications for MRI scan were also excluded. In addition, informed consent was obtained for all volunteers prior to the scan. The average age was 26.7 years (range 20–37) and 14/30 (46.7%) were males. There were 14 right ankles (46.7%) and the side was chosen randomly as long as they fulfilled the inclusion/exclusion criteria.

An isotropic 3D volume acquisition proton-density weighted MRI sequence was used for the study (isotropic 3D-SPACE, 0.5 mm slice thickness, 320 × 320 matrix, ET 40, TR 1200.0 ms, TE 37.0 ms) where volunteers were placed in a 4-channel boot coil with ankle in neutral position during scanning. Images were acquired at 3 Tesla and reformatted from sagittal acquisition to axial and coronal multiplanar reformat images (Siemens Magentom Skyra, Siemens Healthcare HmbH, Erlangen, Germany). MRI scans were independently reviewed by 2 experienced musculoskeletal radiologists (JCL and AWM) with > 20 years of experience with findings being agreed through consensus.

Statistical analysis

The number and size (length and width) of fascicles present in the ATFL, presence of the arciform fibres, anatomical relationship of the ATFL, CFL and PTFL together with the vascular anatomy around the lateral ligamentous complex were assessed. All variables are presented using frequency tables and descriptive statistics (mean, median, range and standard deviation where appropriate). Mann–Whitney U test was used for comparison between continuous variables such as the ATFL fascicle sizes. Values of $p \leq 0.05$ were considered significant. In view of this study being first of its kind, it was designed as a pilot study recruiting 30 healthy volunteers. Data analysis was performed using SPSS (version 16, SPSS Inc., Chicago, IL, USA).

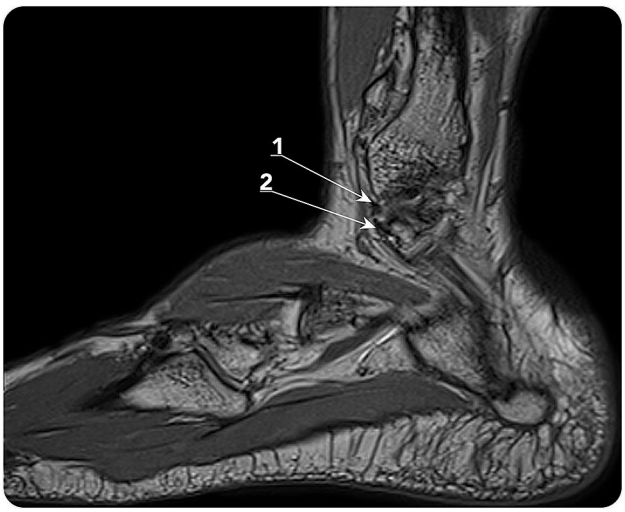


Fig. 2 Sagittal MRI images showing the two ATFL fascicles with (1) superior ATFL fascicle and (2) inferior ATFL fascicle

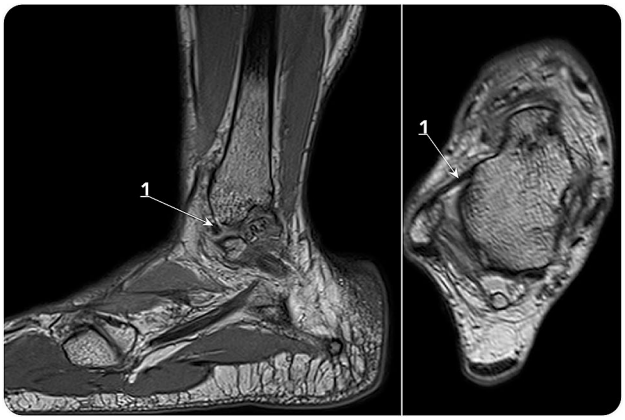


Fig. 3 Sagittal and axial MRI images showing single ATFL fascicle (1)

Results

Two ATFL fascicles were identified in 26 (86.7%) ankles. (Fig. 2) The average length of the superior ATFL fascicle was 18.4 mm (10.8–23.4 mm) and the average width was 4.7 mm (1.8–6.8 mm). In contrast, the inferior ATFL fascicles were significantly smaller with an average length of 13.2 mm (8.0–16.7 mm, $p < 0.001$) and an average width of 2.6 mm (1.4–3.8 mm, $p < 0.001$). The 4 ankles with a single ATFL fascicle had average length of 17.3 mm (range 16.7–18.0 mm) while the average width was 5.9 mm (4.3–7.8 mm) with no significant difference in size when compared to the superior fascicle of those patients in whom two fascicles were identified (length n.s.; width n.s.). (Fig. 3) There were no significant age or gender differences in terms of prevalence of the one or two ATFL fascicles or sizes of the ATFL fascicles. (Table 1).

The LFTCL complex was identified in 3 out of 4 (75%) ankles with single ATFL fascicle while 22 (84.6%) out of 26 of those ankles with two ATFL fascicles had arciform fibres visualized. (Fig. 4) Connecting fibres from the ATFL to PTFL were observed in 19 (63.3%) ankles while connections between the CFL and PTFL were identified in 21 (70%) ankles. (Figs. 5 and 6) A perforating arterial branch from the lateral tarsal artery of the dorsalis pedis artery found in the space between the superior and inferior ATFL fascicles was identified in five ankles. (Fig. 7).

Table 1 Demographics of the 30 ankles with anatomical characteristics of the lateral ligament and lateral fibulotalocalcaneal ligament complexes

Demographics and anatomical characteristics	Number of ankles, $n=30$
Age, mean, (range) years	26.7 (20–37)
Gender, *n* (%)	
Male	14 (46.7%)
Female	16 (53.3%)
Laterality, *n* (%)	
Right	14 (46.7%)
Left	16 (53.3%)
ATFL fascicles, *n* (%)	
Single	4 (13.3%)
Double	26 (86.7%)
ATFL fascicles length, mm (range)	
Single, ($n=4$)	17.3 (16.7–18.0)
Double—superior fascicle, ($n=26$)	18.4 (10.8–23.4)
Double—inferior fascicle, ($n=26$)	13.2 (8.0–16.7)
ATFL fascicles width, mm (range)	
Single, ($n=4$)	5.9 (4.3–7.8)
Double—Superior fascicle, ($n=26$)	4.7 (1.8–6.8)
Double—Inferior fascicle, ($n=26$)	2.6 (1.4–3.8)
Arciform Fibres, *n* (%)	
Single ATFL fascicle, $n=4$	
Yes	3 (75)
No	1 (25)
Double ATFL fascicles, $n=26$	
Yes	22 (84.6)
No	4 (15.4)
ATFL to PTFL, *n* (%)	
Yes	19 (63.3)
No	11 (36.7%)
CFL to PTFL, *n* (%)	
Yes	21 (70)
No	9 (30)
Perforating arterial branch, *n* (%)	
Yes	5 (16.7)
No	25 (83.3)

Discussion

The most important finding of this study is the description and demonstration of the two distinct fascicles of the ATFL with its corresponding sizes on 3D MRI images. In this study, the prevalence of two fascicles of the ATFL in MRI ankles of healthy volunteers was 86.7%. However, literature on prevalence of ATFL fascicles in MRI studies are sparse with only two recent publications where Choo et al. reported a prevalence of 81.8% (27 out of 33 volunteers) for double banded ATFL in MRI scans of healthy volunteers [2]. Szaro et al. reported prevalence of 74.7%



Fig. 4 Sagittal MRI image of (1) inferior fascicle of ATFL connected by (2) arciform fibres to the (3) CFL forming the LFTCL complex

(148 out of 198 patients) for the presence of double fascicular ATFL in MRI scans although there was no mention about whether these were healthy volunteers or patients treated for other foot and ankle pathology [13]. An older MRI study in 2003 by Delfaut et al. reported 55% (12 out of 22) bifascicular appearance of the ATFL, 36% (8 out of 22) striated appearance while only 9% (2 out of 22) had monofasciculated appearance [5]. They did not use T1 sequences or fat saturated sequences in their study and the thicker slices of 3.0 mm may have meant that the bifascicular nature of ATFL could have been missed [5]. Overall, reports in the literature point that two ATFL fascicles may be identified on MRI scans in at least 75% of individuals [2, 12, 13]. The absence of two separate fascicles in some patients may support Vega et al. suggestion that a single ATFL fascicle is pathological and secondary to previous sprains resulting in isolated injury of the superior ATFL fascicle and subsequent resorption leading to a hypertrophied inferior ATFL fascicle masquerading as a single fascicular ATFL [18]. We also found that the superior fascicle is significantly larger than the inferior fascicle of the ATFL in all specimens at 39% longer and 80.7% wider on average. None of the inferior fascicles were larger than the superior fascicles while the single ATFL fascicle had almost similar size to the superior fascicle of the ATFL. These finding were consistent with other MRI studies [2, 5, 13]. Clinically, it is vital to be able to identify these individual fascicles of the ATFL as recent studies have suggested that isolated injury of the superior fascicle can

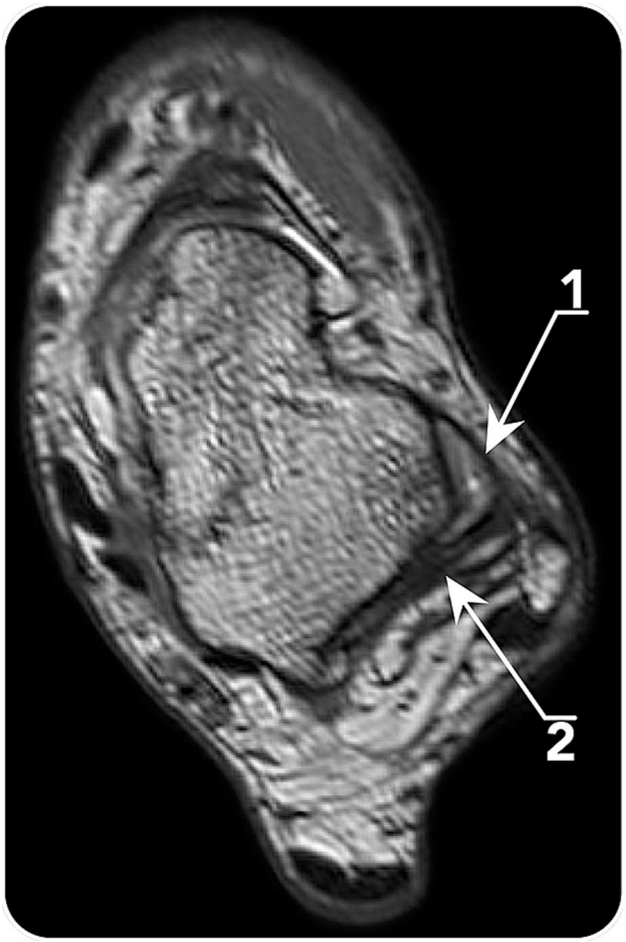


Fig. 5 Axial MRI image showing (1) ATFL connecting to (2) PTFL

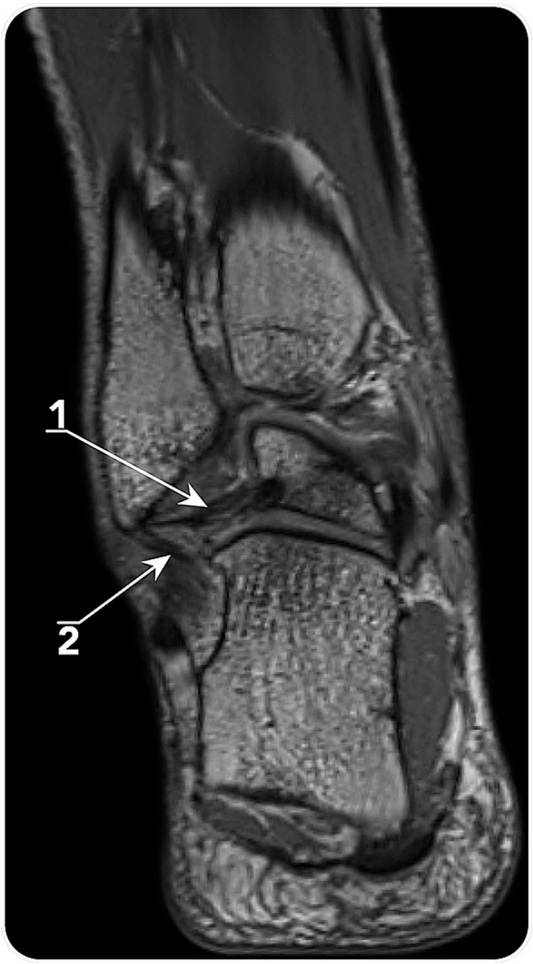


Fig. 6 Coronal MRI image showing (1) CFL connecting to (2) PTFL

lead to subtle instability and chronic anterolateral ankle pain which is known as "microinstability" [3, 16–19]. Injuries of the superior fascicle of ATFL may have a poor healing potential due to its intra-articular nature whereby synovial fluid bathes the ligament, similar to other intra-articular ligaments [11, 15, 17, 18]. Therefore, we posit that fascicle-specific ATFL injuries can be diagnosed on MRI and that isolated superior ATFL fascicle injury could potentially be identified routinely in MRI as a cause for symptoms related to "microinstability" [16–18].

Another important finding of this study is the identification of the arciform fibres connecting the inferior ATFL fascicle and CFL to form the LFTCL complex. A recent study looked at MRI images of 198 ankles retrospectively and found connections between ATFL and CFL in 74.7% [13]. However, they have only used proton density (PD) or T2 sequences with thicker slicing and therefore were not able to provide clear description or illustrations of the fibres [13]. Choo et al. evaluated the MRI characteristics of multi banded ATFL using 3D MRI [2]. They reported the number of bundles, but they did not look for connections between

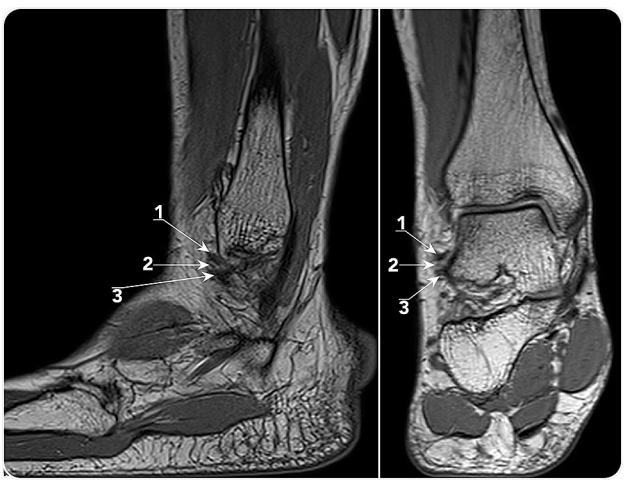


Fig. 7 Sagittal MRI image showing the (1) superior fascicle of ATFL and (3) inferior fascicle of ATFL with the intervening perforating artery (2). Coronal MRI image illustrating the (2) perforating artery from lateral tarsal branch of the dorsalis pedis artery in the space between the (1) superior fascicle of ATFL and (3) inferior fascicle of ATFL

ATFL and CFL [2]. Nakasa et al. reviewed 48 preoperative MRI scans of the ankle with 0.8 mm slices in 45 patients who underwent arthroscopic surgery and defined injury of the LFTCL complex when either the inferior fascicle of ATFL or CFL was observed to be injured on MRI [12]. Despite labelling the LFTCL complex, once again they did not specifically identify or demonstrate the arciform fibres [12]. The current study demonstrates for the first time on MRI images, the arciform fibres which formed the anatomical and functional structure of the LFTCL complex. This is important because arciform fibres is not just an anatomical structure connecting the inferior ATFL fascicle to the CFL, but it has been shown to be functionally strong enough to transmit tension from one structure to the other, therefore supporting isolated ATFL repair to restore ankle stability in the event of injuries to both CFL and ATFL [3, 9, 10, 18]. Several studies have reported long term outcomes of isolated ATFL repair with excellent results [9, 10] which further lend its support to the notion that repairing only the ATFL is in fact already a truly anatomical repair as it indirectly transmits tension to the CFL via arciform fibres [3, 18]. The only caveat to this is that the CFL injury must occur more proximally to preserve the common connecting arciform fibres [3]. The clinical implications that follow visualization of this LFTCL complex can affect surgical treatment: the loss of arciform fibres seen in MRI could potentially lead to poorer outcomes in isolated ATFL repair due to the loss of connection to the CFL and perhaps a combined ATFL and CFL repair is necessary in the case of LFTCL complex injuries.

Gosselin et al. studied the vascular supply to the lateral ankle ligamentous complex in 56 cadaveric specimens and found that the ATFL in all specimens had robust blood supply with contributions from more than one source [8]. They demonstrated that the lateral tarsal artery branching from the dorsalis pedis supplied 29 (51.8%) specimens while the anterior perforating branch from peroneal artery supplied 52 (92.9%) specimens [8]. Although Gosselin and colleagues did not identify the perforating branch in between the two fascicles of the ATFL, they provided evidence that the ATFL has excellent perfusion from two main arteries [8]. Golano et al. [7] and Vega et al. [18] reported a vascular branch from the peroneal artery found in the gap between the two fascicles of the ATFL. Interestingly, we have only managed to identify the perforating arterial branch from the lateral tarsal artery of the dorsalis pedis artery piercing into the space between the superior and inferior ATFL fascicles in 5 ankles. The clinical relevance of this finding is twofold: (1) the perforating arterial branch can be used as a landmark to distinguish the superior and inferior fascicles of the ATFL and (2) it is likely to be the cause for the extensive haematoma seen in some patients following an ankle sprain. Furthermore, we were also able to identify and characterize connecting fibres from ATFL to PTFL and CFL to PTFL in the MRI images. Up to two thirds (63.3%) of the ankles have connecting fibres from ATFL to PTFL while CFL to PTFL connections were noted in 70% of the ankles. This is consistent with a cadaveric dissection study by Dalmau-Pastor et al. where they found interconnections amongst the individual structures of the lateral ankle ligaments from the medial aspect of the ankle [4].

There are several limitations in this study. Although all the volunteers were healthy and asymptomatic with no ankle instability symptoms in the preceding 6 months at the point of recruitment, there is a potential criticism that they may have had previous ankle sprains which had healed over time. This could confound certain findings such as the superior ATFL fascicle that could be ruptured and atrophied with hypertrophy of the inferior ATFL fascicle as compensatory mechanism. In addition, even though these images were obtained using 3 T MRI scanner with 0.5 mm slice thickness to improve accuracy of anatomical structures identification, coupled with image evaluation by two very experienced musculoskeletal radiologists independently, there were still risks of missing very small/thin inferior ATFL fascicles, arciform fibres or even connecting fibres from ATFL/CFL to PTFL. Finally, 3 T MRI scanners may not be a commonly available resource. Despite these limitations, to the best of our knowledge, this study is the first to describe and demonstrate the arciform fibres as part of the LFTCL complex in the ankle on MRI scans. As a result of this study, fascicle-specific ATFL injuries can be identified on MRI routinely so that isolated superior ATFL fascicle injury could be potentially diagnosed as a cause for subtle lateral ankle instability. In terms of identifying the arciform fibres of LFTCL complex, isolated ATFL repair can be attempted with good outcomes in the presence of intact LFTCL complex.

Conclusion

Majority of ATFL have two distinct fascicles on MRI imaging and fascicle-specific injury such as that to the superior ATFL fascicle can be distinguished to diagnose subtle lateral ankle instability. The arciform fibres as part of the LFTCL complex have been described in this current study for the first time on MRI scans and may play an important role in deciding on surgical treatment with an isolated ATFL repair.

Author contributions All authors contributed to the study conception and design. AT, MK and MJ: prepared the study protocol and performed the ethical review process. Material preparation, data collection and analysis were performed by CCH, JCL, AWM, MD-P and JC. The first draft of the manuscript was written by CCH and all authors commented on previous versions of the manuscript. JC: supervised and reviewed the final manuscript. All authors read and approved the final manuscript.

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Data availability statement Authors can confirm that all relevant data are included in the article and/or its supplementary information files.

Declarations

Conflict of interest All authors do not have any conflict of interest except for James Calder who received remuneration for speaking in educational programme by Arthrex unrelated to this study.

Ethical approval Institutional approval to undertake this study was provided by the West Midlands—Coventry & Warwickshire Research Ethics Committee (Reference: 19/WM/0383).

Informed consent Informed consent was obtained for all volunteers prior to the scan.

References

1. Broström L (1964) Sprained ankles. I. Anatomic lesions in recent sprains. Acta Chir Scand 128:483–495
2. Choo HJ, Lee SJ, Kim DW, Jeong HW, Gwak H (2014) Multibanded anterior talofibular ligaments in normal ankles and sprained ankles using 3D isotropic proton density-weighted fast spin-echo MRI sequence. AJR Am J Roentgenol 202(1):W87-94
3. Cordier G, Nunes GA, Vega J, Roure F, Dalmau-Pastor M (2021) Connecting fibers between ATFL's inferior fascicle and CFL transmit tension between both ligaments. Knee Surg Sports Traumatol Arthrosc 29(8):2511–2516
4. Dalmau-Pastor M, Malagelada F, Calder J, Manzanares MC, Vega J (2020) The lateral ankle ligaments are interconnected: the medial connecting fibres between the anterior talofibular, calcaneofibular and posterior talofibular ligaments. Knee Surg Sports Traumatol Arthrosc 28(1):34–39
5. Delfaut EM, Demondion X, Boutry N, Cotten H, Mestdagh H, Cotten A (2003) Multi-fasciculated anterior talo-fibular ligament: reassessment of normal findings. Eur Radiol 13(8):1836–1842
6. Garrick JG (1977) The frequency of injury, mechanism of injury, and epidemiology of ankle sprains. Am J Sports Med 5(6):241–242
7. Golanó P, Vega J, de Leeuw PA, Malagelada F, Manzanares MC, Götzens V, van Dijk CN (2016) Anatomy of the ankle ligaments: a pictorial essay. Knee Surg Sports Traumatol Arthrosc 24(4):944–956
8. Gosselin MM, Haynes JA, McCormick JJ, Johnson JE, Klein SE (2019) The arterial anatomy of the lateral ligament complex of the ankle: a cadaveric study. Am J Sports Med 47(1):138–143
9. Lee KT, Park YU, Kim JS, Kim JB, Kim KC, Kang SK (2011) Long-term results after modified Brostrom procedure without calcaneofibular ligament reconstruction. Foot Ankle Int 32(2):153–157
10. Maffulli N, Del Buono A, Maffulli GD, Oliva F, Testa V, Capasso G, Denaro V (2013) Isolated anterior talofibular ligament Broström repair for chronic lateral ankle instability: 9-year follow-up. Am J Sports Med 41(4):858–864
11. Murray MM, Spindler KP, Ballard P, Welch TP, Zurakowski D, Nanney LB (2007) Enhanced histologic repair in a central wound in the anterior cruciate ligament with a collagen-platelet-rich plasma scaffold. J Orthop Res 25(8):1007–1017
12. Nakasa T, Ikuta Y, Sumii J, Nekomoto A, Kawabata S, Adachi N (2022) MRI appearance of the lateral fibulotalocalcaneal ligament complex injury in the patients with chronic lateral ankle instability. Foot Ankle Surg 28(7):968–974
13. Szaro P, Ghali Gataa K, Polaczek M, Ciszek B (2020) The double fascicular variations of the anterior talofibular ligament and the calcaneofibular ligament correlate with interconnections between lateral ankle structures revealed on magnetic resonance imaging. Sci Rep 10(1):20801
14. van den Bekerom MP, Kerkhoffs GM, McCollum GA, Calder JD, van Dijk CN (2013) Management of acute lateral ankle ligament injury in the athlete. Knee Surg Sports Traumatol Arthrosc 21(6):1390–1395
15. Vavken P, Murray MM (2011) The potential for primary repair of the ACL. Sports Med Arthrosc Rev 19(1):44–49
16. Vega J, Dalmau-Pastor M, Malagelada F, Fargues-Polo B, Peña F (2017) Ankle arthroscopy: an update. J Bone Joint Surg Am 99(16):1395–1407
17. Vega J, Malagelada F, Dalmau-Pastor M (2021) Ankle microinstability: arthroscopic findings reveal four types of lesion to the anterior talofibular ligament's superior fascicle. Knee Surg Sports Traumatol Arthrosc 29(4):1294–1303
18. Vega J, Malagelada F, Manzanares Céspedes MC, Dalmau-Pastor M (2020) The lateral fibulotalocalcaneal ligament complex: an ankle stabilizing isometric structure. Knee Surg Sports Traumatol Arthrosc 28(1):8–17
19. Vega J, Peña F, Golanó P (2016) Minor or occult ankle instability as a cause of anterolateral pain after ankle sprain. Knee Surg Sports Traumatol Arthrosc 24(4):1116–1123
20. Waterman BR, Belmont PJ Jr, Cameron KL, Deberardino TM, Owens BD (2010) Epidemiology of ankle sprain at the United States Military Academy. Am J Sports Med 38(4):797–803
21. Woods C, Hawkins R, Hulse M, Hodson A (2003) The Football Association Medical Research Programme: an audit of injuries in professional football: an analysis of ankle sprains. Br J Sports Med 37(3):233–238

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